

Performance Testing

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Custom Python load script <code>tests/perf_load.py</code> (httpx + threads, psutil for CPU/RAM sampling); functional tests with pytest (<code>tests/test_app.py</code>)
Type of Testing	Load testing (1, 10 and 25 concurrent virtual users, 20 s each) and functional regression
Target Module / API	GET <code>/</code> , GET <code>/health</code> and POST <code>/qa</code> with empty text (rejected by validation before any Gemini call)
Test Environment	Linux cloud machine (1 vCPU, about 4 GB RAM, Python 3.12.3), single Uvicorn worker, load generator on the same machine. Gemini is deliberately not called, so the results measure EduGenie's own overhead and use no API quota
Test Date	08 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline: one user cycling through the three requests	1	20	Fast responses, 0 errors
2	Typical classroom load	10	20	Avg < 2 s, 0 errors
3	Peak load	25	20	Avg < 2 s, max < 5 s, error rate < 1 %
4	Functional regression under test (pytest)	1	-	All tests pass

Step 3: Performance Test Results (25 virtual users)

Measured by running `tests/perf_load.py` for 20 seconds with 25 virtual users (the raw output is in the evidence folder).

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.022 seconds	Pass	95th percentile 0.029 s
2	Response Time (Max)	< 5 seconds	0.177 seconds	Pass	Worst single request in 22,642
3	Throughput (Req/sec)	Not specified	1,130.8 req/s	N/A (no target set)	22,642 requests in 20 s

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
4	Error Rate	< 1%	0.00 %	Pass	No failed requests
5	CPU Utilization	< 80%	42 % average (44 % peak) of one core	Pass	Shared with the load generator on the same single core, so the real server figure is lower
6	Memory Utilization	< 80%	67 MB (about 1.7 % of 4 GB RAM)	Pass	Steady throughout the run

Results at each load level

Virtual Users	Requests	Throughput (req/s)	Avg Response (s)	95th Percentile (s)	Max Response (s)	Error Rate	CPU Avg	Memory
1	22,010	1,099.9	0.001	0.001	0.007	0.00 %	48 %	67 MB
10	24,025	1,200.2	0.008	0.011	0.034	0.00 %	43 %	67 MB
25	22,642	1,130.8	0.022	0.029	0.177	0.00 %	42 %	67 MB

Functional tests (pytest)

32 tests ran and **32 passed** (0 failed) in 0.11 seconds.

Step 4: Observations & Analysis

- All six metrics met their targets at 25 virtual users: average response 0.022 s (target < 2 s), maximum 0.177 s (target < 5 s) and 0 % errors (target < 1 %).
- Response time grows gently with load (0.001 s with 1 user, 0.008 s with 10 users, 0.022 s with 25 users) while throughput stays near 1,100 to 1,200 requests per second. The server is not saturated at this load.
- CPU stayed around 42 to 48 % of one core and memory stayed at about 67 MB at every load level, so EduGenie has plenty of headroom for a classroom-sized group.
- In normal use the response time is dominated by the Gemini call (typically a few seconds), not by EduGenie, which only validates the input and builds a prompt.
- The load test avoids Gemini on purpose: it keeps results repeatable and does not consume the free-tier quota. As a result, live Gemini latency was **not** measured by this test.
- The test used a single machine for both the server and the load generator, so absolute numbers will differ on other hardware. The commands in Step 5 can be re-run on any laptop.
- Scalability next steps: several Uvicorn workers, response caching and rate limiting (see Scalability & Future Plan).

Step 5: Screenshots / Evidence

The raw output of every run is saved in the 6.Project Testing/evidence folder:

File	Contents
load_1_users.txt	Load test, 1 virtual user, 20 s
load_10_users.txt	Load test, 10 virtual users, 20 s
load_25_users.txt	Load test, 25 virtual users, 20 s

File	Contents
pytest_results.txt	Functional tests, 32 passed

Output of the 25-user run:

```
virtual users      : 25
duration (s)      : 20.0
requests          : 22642
throughput (req/s) : 1130.8
avg response (s)  : 0.022
p95 response (s)  : 0.029
max response (s)  : 0.177
error rate (%)     : 0.00
server CPU avg/max : 42% / 44% (of one core)
server RSS avg/max : 67 MB / 67 MB
```

Output of the functional tests (last line): 32 passed, 1 warning in 0.11s

To repeat the tests, run these from the project root:

```
uvicorn main:app --port 8000 # terminal 1
python tests/perf_load.py http://localhost:8000 1 20 > "6.Project Testing/evidence/load_1_
python tests/perf_load.py http://localhost:8000 10 20 > "6.Project Testing/evidence/load_10_
python tests/perf_load.py http://localhost:8000 25 20 > "6.Project Testing/evidence/load_25_
pytest tests -v > "6.Project Testing/evidence/pytest_results.txt"
```

Functional tests cover: a valid question, follow-up context, empty / invalid / too-long input on every endpoint, summary, quiz generation (including code fences and bad Gemini output), Gemini failure with hidden details, empty Gemini reply, missing key, rejected key, model fallback, rate limit, hidden unexpected errors, and the page-to-API round trip.